MATH 20250
Prof. Lanier

Handout, Day 2

March 22, 2023

Warm-up 1

Compute in $\mathbb{R}^2$:

$$\begin{pmatrix} 2 \\ -1 \end{pmatrix} + 5 \begin{pmatrix} 3 \\ 4 \end{pmatrix} =$$

Find scalars $a, b \in \mathbb{R}$ that make the equality true, or explain why this is not possible.

$$a \begin{pmatrix} 5 \\ 2 \end{pmatrix} + b \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Warm-up 2

Recall the "avoiding square products" game from yesterday's warm-up. Here is a set of 10 numbers. Is there a (non-empty) subset of them whose product is a square number? If so, what is this subset? If not, how many numbers must be removed from the set so that no square products remain?

$$\{2, 6, 7, 10, 11, 15, 18, 19, 22, 23\}$$